UCC TESTBED HACKATHON 2026 / TRACK 4

Product Requirements Document

AI-Powered Child Cyber Safety, Prevention, Support and Threat Intelligence

PROJECT
Chinga (working name)

VERSION
2.0 — Revised scope

DATE
5 November 2026

PRIMARY CHALLENGE
Child-Focused Cybersecurity and Online Fraud Prevention

PRODUCT STRATEGY · IMPLEMENTATION SCOPE · SAFEGUARDING

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System vision, Android protection capabilities, guardian links, child support, threat intelligence, responsible AI and hackathon delivery.

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1. Executive summary

Chinga is an AI-powered digital-safety platform designed for children and families in Uganda. It combines **fraud prevention**, **website risk assessment**, **age-appropriate browsing protection**, **confidential child-friendly assistance**, **guardian support**, and **scam-pattern intelligence**.

The product is deliberately split into two layers. First, an **Android child-protection companion** supports device-level controls to the extent permitted by Android, while a **mobile-friendly web/PWA interface** provides universally accessible checking, guidance and parent services. Second, a shared **backend and AI engine** analyses text, URLs and screenshots; supports evidence-based reporting; and produces a carefully de-identified intelligence feed for an authorised administrator dashboard.

The two-day hackathon deliverable is a functional prototype: child-to-parent pairing; real scam/link analysis on the assigned UCC Testbed; an Android or PWA link-check flow; a simple child chat/reporting interface; a Sauti 116 referral button; and a privacy-protected dashboard. Android-wide domain filtering, live voice, advanced parental controls, and fully operational SMS/Telegram are important extensions, but must only be marked **live** when demonstrated end-to-end.

2. Problem and opportunity

Ugandan children encounter phishing, fraudulent scholarship claims, fake Mobile Money rewards, impersonation, malicious links, and potentially age-inappropriate online content. They often lack the knowledge, confidence, mobile data or trusted support to evaluate a suspicious message or explain what happened after an incident.

Parents want to protect children without becoming permanent readers of every private conversation. Meanwhile, authorised child-protection and cybersecurity organisations need **actionable scam evidence and patterns**, not just an undifferentiated pile of reports. Chinga addresses these needs through prevention, child-friendly support, parent-configured protection and aggregated intelligence.

This project is primarily **Track 4 cybersecurity/fraud prevention**. Its reporting and welfare features support that core use case; they are not presented as a substitute for clinical mental-health treatment or official child-protection case management.

3. Product goals and non-goals

3.1 Goals

1. Help a child assess **text, links, screenshots, and shared content** before clicking, paying, or disclosing sensitive information.
2. Warn or block **known unsafe domains and disallowed content categories**, within the technical scope of the configured protection method.
3. Give age-appropriate, understandable reasons for each decision, including an **uncertain** outcome when evidence is insufficient.
4. Link a guardian and child through a verified, expiring pairing mechanism and deliver **appropriate, minimal** safety alerts.

5. Offer a private, compassionate AI conversation for online safety questions and optional incident reporting.
6. Provide a visible, immediate pathway to **Uganda Sauti 116** and other approved support services.
7. Support English initially; plan and evaluate **Luganda and Runyankole** rather than claiming untested accuracy.
8. Provide a usable **UCC-facing threat intelligence dashboard** with sanitized scam-message excerpts, impersonated organisations, patterns, trends and controlled exports.
9. Share the same security service across Android, PWA, Telegram, SMS and future voice interfaces.
10. Demonstrate actual use of UCC Testbed infrastructure and report meaningful evaluation metrics.

3.2 Non-goals

- Reading entire WhatsApp, Telegram or SMS conversations without user action and valid OS permissions.
- Guaranteed interception of **every** click or page on an unmanaged Android phone.
- Secret surveillance, indefinite collection of a child's browsing history, or unrestricted parent access to private disclosures.
- Claiming that an impersonated institution caused a scam or suffered a breach.
- Automatic police/helpline contact, diagnosis, or suicide-risk determination solely from an LLM verdict.
- Circumventing Android controls or installing the child app as a privileged system app without legitimate device enrollment.

4. User roles

Role	Primary capabilities	Key restrictions
Child	Check links and messages, share screenshots, seek advice, report experiences, call 116	Can use basic help without being forced into a guardian account; privacy respected
Parent / guardian	Register, pair children, set age-appropriate rules, review limited safety alerts, ask the parent AI for advice	Cannot automatically access full child chats or confidential reports
Authorised UCC/admin analyst	View de-identified incident trends, sanitised scammer evidence, campaign links, institution impersonation and exports	Role-restricted evidence access; no indiscriminate child-identifying data
Safeguarding reviewer	Review escalated safety cases under formal safeguarding rules	Human oversight, logging, minimum necessary information
System operator	Manage models, integrations, policies and security/audit logs	No unrestricted user content access by default

5. Supported product surfaces

Surface	Purpose	MVP decision
Android child-protection app	Native VPN/domain filtering pilot, safe-link flow, permission settings, device heartbeat	**Prototype one working pathway**, not full device control
Next.js PWA	Child checking, guardian registration, web chat, reports, notifications	**Core deliverable**
Parent portal (PWA/mobile)	Child pairing, rules, alerts, parent AI advice	**Core basic flows**
UCC web dashboard	Scam evidence, campaign grouping, metrics, reports and export	**Core deliverable**
Telegram bot	Message/link/screenshot checking through shared API	Scaffold or implement if time permits
SMS / Africa's Talking	No-data text checking and guided reporting	Provider-ready interface; sandbox if feasible
Voice AI	Live voice input/output with image submissions and child-friendly conversation	Later phase / stretch prototype
Browser extension	Check link context in a supported browser before opening	Later phase; optional desktop demonstration
WhatsApp Business channel	Chat with assistant through WhatsApp	Future integration, not required for MVP

6. Child and guardian onboarding

6.1 Parent enrollment

1. Select **I am a parent/guardian**.
2. Provide a phone number and complete OTP verification where available.
3. Create a parent profile with minimal required information.
4. Add a child **nickname** and age band; do not require the child's full legal name for routine checking.
5. Review clearly explained permissions and privacy choices.
6. Generate an **expiring, single-use linking code**.
7. After child pairing, configure browsing categories and safety-notification preferences.

6.2 Child enrollment

1. Select **I have a family code** or **Get help now**.
2. Enter the linking code on the child's device, if applicable.

3. Confirm the child account has been securely paired and display what information guardians can see.
4. Offer the same two main actions: **Is This Safe?** and **Something Happened**.
5. Show **Talk to someone** / **Call 116** as a third visible support path, without overwhelming the interface.

A child who cannot safely involve a parent should still be able to obtain basic advice and access help. Guardian verification, legal basis and child safeguarding processes require review before public deployment.

7. Child experience and conversational assistant

7.1 'Is This Safe?' flow

Accept typed/pasted text, URLs, screenshots and content shared into the app. The assistant extracts relevant scammer-supplied content from screenshots, identifies suspicious claims, checks technical signals and replies in simple language.

Response states: **Low observed risk**, **Suspicious**, **High risk**, **Unable to verify**. Never label a site '100% safe'. Provide a short reason and a next step. If the child has already clicked or shared a credential, switch to the incident-support flow automatically.

7.2 'Something Happened' flow

The AI can ask a small number of focused questions about a prior scam, financial loss, shared credentials or threatening interaction. It must never ask for PINs, passwords or OTPs. It builds a summary the child can correct before deciding whether to submit a formal report.

7.3 Confidential 'Talk to someone' flow

Offer an emotionally supportive online-safety conversation for children who do not want face-to-face help. The assistant should listen, avoid judgement, and give practical next steps, including referral to trained human services. It is **not** a therapist or replacement for safeguarding professionals.

If the child describes serious immediate danger, abuse, or thoughts of suicide/self-harm, the system follows a dedicated **high-risk support policy**: respond empathetically, encourage immediate contact with a trusted and safe adult and emergency/helpline services, offer direct access to 116, and refer to human support where approved. Do **not** send raw private conversations or automatically notify a potentially unsafe guardian. A trained safeguarding procedure determines the narrow circumstances and minimum information for urgent escalation.

7.4 Parent AI advisor

Parent accounts can ask about suspected scams, age-appropriate internet rules, handling online harm, and how to speak with a child. Responses should teach and guide; they should not fabricate access to confidential child chats.

8. Android protective controls: what is actually possible

The Android companion is a **separate native capability layer** connected to the same backend. Permission grants are **feature-specific** and never amount to unrestricted control of the device.

Feature	Android technique	What it can and cannot do
Check a link before opening	Make Chinga a default browser/ handler for supported web intents, or receive a link via Share	Can inspect complete URLs that actually reach Chinga; cannot guarantee that all third-party in-app browsers pass through it
Block domains across many apps	User-approved `VpnService` with a tested DNS/network filtering pipeline	Can block many known domains; generally cannot read full encrypted URL paths or all page contents; only one VPN active per Android user
Filter gambling/adult/scam domains	Verified policy lists, DNS/domain signals, optional threat lookup	Coverage depends on DNS/protocol visibility, browser behavior, VPN state and bypass resistance; educational/health resources need reviewable exceptions
App usage trends	Android Usage Access where permitted	Some usage data possible after user action; does not grant private chat access
Hide or disable arbitrary apps	`DevicePolicyManager` under eligible device-owner/profile-owner management	**Not available merely because a parent taps 'Allow all permissions'**; managed-device enrollment and OS policy are required
Detect app removal	Missed server check-ins; managed-device status where available	Unmanaged app cannot reliably report its own deletion after removal; missing check-ins are not proof of uninstall
Alert if protection stops	Heartbeats, foreground service/VPN state where supported	Network outages, battery policies and device shutdown may cause missing signals; avoid presenting them as definite tampering

8.1 Safe web-navigation sequence

1. Child taps a link that is actually routed to Chinga, **or** the phone attempts a network request while Chinga's approved VPN filter is active.
2. Exact URL analysis is used only when the full URL is available; network-level checks use the domain or other visible connection metadata.
3. The policy engine checks known malicious domains, suspicious lookalikes, child-content category, parent settings, and approved exceptions.
4. If strong evidence or policy justifies blocking, the app stops navigation **within its available control path** and explains why.
5. If risk is ambiguous, show **Review / Ask for help**, rather than giving an unjustified absolute guarantee.

6. Log only minimal event metadata for analytics and parent alerts; avoid saving full browsing histories by default.

8.2 Guardian control policy

Parents may set allowed/disallowed site categories, bedtime/usage preferences and risk alerts, subject to device capability and age-appropriate choice. Restrictions must not block critical help resources, including 116 or emergency services, and must not be presented as sufficient to prevent every form of harm.

An optional 'guardian mode' dashboard may display policy state and last successful check-in. Any **daily activity** display must mean 'last observed app/device protection check-in', not a definitive log of every time the child used the phone.

8.3 Android privacy and deployment rules

VPN activity requires the Android system's explicit consent flow, a visible protection state, minimal necessary inspection, and compliance with Google Play VPN/parental-control policies. Never enable Accessibility services to secretly read other apps. Device-owner features are a future managed-device track and require separate eligibility and testing.

9. AI engine and evidence-based risk assessment

9.1 Shared analysis pipeline

Input -> text/vision extraction -> URL and claimed-entity parsing -> technical checks -> trained/pretrained classifier(s) -> contextual LLM reasoning -> policy decision -> child-friendly explanation -> minimal evidence record.

The vision-capable LLM may extract screenshot content and sender roles. When the sender is unclear, ask the child which message was received rather than guessing. The LLM may explain and contextualise evidence, but **its unsupported assertion must not override reliable technical signals**. On disagreement, request more evidence or return 'Unable to verify'.

9.2 Security signals

- Known-bad matches from PhishTank / URLhaus / other approved feeds.
- URL lexical and lookalike-domain features (including Unicode and zero/O confusables).
- Claimed organisation versus verified official-domain mismatch.
- Requests for OTP, PIN, password, bank details or advance payment.
- Fake prizes, urgent deadlines, impersonated schools/scholarship programmes.
- Domain/category filtering according to guardian preferences and current policy.
- Similarity to previously reviewed scam submissions, with duplicate detection.

Avoid rendering or executing suspect web content unsafely. Do not automatically browse arbitrary untrusted links from the AI's server without isolation and explicit safeguards.

9.3 Decision data contract

```
{
  "analysisId": "example-001",
  "mode": "live-or-demo",
  "riskLevel": "suspicious",
  "category": "phishing",
  "signals": ["claimed-brand-domain-mismatch"],
  "evidenceSources": ["url-check", "message-classifier"],
  "explanation": "This link may be pretending to represent a trusted company.",
  "recommendedActions": ["Do not provide your PIN."],
  "needsHumanReview": false,
  "relatedCampaignId": null
}
```

A risk score, if included, must be labelled as an **uncalibrated model score** unless validated. Every result must identify whether it is live inference, a fixture-based simulation, or unable to verify.

10. Website age-suitability policy

Provide configurable categories such as **fraud/phishing**, **gambling**, **explicit adult content**, **malware** and additional locally relevant risks. For a child-focused safety product, blocking is a **policy decision** based on reviewed rules and reliable categorisation, not an unreviewed LLM opinion alone.

The classifier can consider page/domain metadata and curated categorisation feeds when technically and legally available. False positives should be reviewed, especially for sexual-health education, legitimate news, educational references, and support services. Provide a correction / appeal pathway for parents and authorised reviewers.

11. Scam Wave Intelligence

Scam Wave Intelligence connects separate reports into suspected scam campaigns using message meaning, copied templates, suspicious domains, requested actions, impersonated entities, timestamps and district-level trends. It must distinguish **submission count** from confirmed unique victims, and avoid treating duplicate forwards as independent incidents.

The dashboard may show a *suspected fake MTN reward campaign*, related sanitised scammer messages, common tactics, linked domains and trend changes. It must not assert that MTN - or any impersonated institution - organised the scam or suffered a breach.

For the hackathon, rule-based grouping or prepared synthetic campaign fixtures are acceptable **if accurately labelled**. Model-powered clustering must be demonstrated before claiming it is running.

12. UCC administrator dashboard

Core views:

1. **Overview:** submissions, flagged links, incident reports, review status and time trends.
2. **Threat evidence:** relevant sanitised scammer text, visible risk signals, observed domain and classification rationale.
3. **Scam Wave Intelligence:** suspected campaigns, matching patterns, evidence examples and locations at safe aggregation levels.
4. **Impersonated organisations:** count and type of impersonations, clearly marked 'alleged impersonation' rather than institutional fault.
5. **Incident and safeguarding review:** status, assigned reviewer and restricted evidence access.
6. **Model evaluation:** measured precision/recall, false positives/false negatives, languages and failure categories.
7. **Exports:** CSV and PDF summaries with authorisation, redaction and audit logging. Formal referrals require an approved workflow.

No child-identifying information or full private conversation should appear by default in overview panels or ordinary exports. Only role-authorised human reviewers can access additional evidence when justified. Use synthetic/demo data for the hackathon.

13. Confidential reporting, permissions and escalation

- **Routine check:** no automatic disclosure of private chat text to parents or admins.
- **Optional report:** let the child review what is included and what will be shared; provide plain-language information about confidentiality and its limits.
- **Guardian alert:** communicate the minimum needed to address a material cybersecurity risk, if guardian contact is appropriate and safe under policy.
- **Safeguarding concern:** follow a trained human-reviewed escalation procedure and relevant law; do not assume all under-18 conversations are automatically disclosed to guardians.
- **Evidence handling:** store only relevant scammer-supplied excerpts and minimal incident metadata, redact secrets and child identifiers, apply encryption, role-based access and retention rules.
- **Official case submission:** do not imply that UCC, police, schools or other bodies have received a report until their actual submission endpoint or authorised manual process confirms receipt.

The first release must have clear consent, safeguarding review and an operational procedure before any automatic third-party notifications are activated.

14. Uganda Sauti 116 and referral workflows

Sauti 116 is Uganda's toll-free child helpline. Provide a prominent **Call 116** action, clear text explaining when to use it, and help for children who prefer to talk with an adult. The call action should hand control to the phone dialer; it must not pretend to have completed a call.

Potential future integration: with a formal agreement from the helpline operator, route selected reports or request a call-back using authenticated, reviewed procedures. **No automated outbound incident transmission to 116, law enforcement or third-party organisations** without approved integration, a legally appropriate basis and a human-in-the-loop safety design.

In urgent safety situations, the assistant must encourage reaching a safe adult or appropriate emergency support promptly rather than promising continuous real-time monitoring.

15. Multilingual and voice support

Target languages: **English, Luganda and Runyankole**. Provide language selection, a simple fallback to English, and explicit limitations when speech recognition or response accuracy is uncertain. Evaluate safety terminology and scam detection in each language; do not equate 'LLM understands some Luganda' with clinically or legally reliable support.

A later voice interface can use speech-to-text -> shared AI agent -> text-to-speech, with an optional screenshot upload during the voice session. Seek child-appropriate consent and minimise retention of audio. Voice must not remove the accessible text interface.

16. Telegram, SMS and future WhatsApp

Telegram: bot receives voluntarily submitted text, links and images; securely associates an account when opted in; sends the shared engine's safety advice. Do not claim access to users' unrelated Telegram conversations.

SMS: optional provider (e.g. Africa's Talking) accepts user-sent messages, sends concise warnings, supports short reporting sessions and applies cost/rate controls. Toll-free delivery needs a genuine operator agreement; do not describe it as already funded.

WhatsApp: future official Business Platform integration. A user can alternatively share content into Chinga where supported by the Android share mechanism. An installed PWA's share target works only under supported browser/platform conditions.

17. Device health, check-ins and parental notifications

When Android protection is enabled, the app can send periodic **service heartbeat** events as permitted by operating-system scheduling, battery restrictions and network access. The guardian dashboard can show:

- Last confirmed check-in.
- Last observed protection/VPN status.
- Alert when protection has not checked in for an agreed interval.
- The distinction between **offline, disabled, inaccessible and uninstalled** being unknown without additional managed-device evidence.

No continuous tracking guarantee is implied. Do not implement coercive or deceptive reinstallation, silent surveillance or inescapable lockdowns. Escalation and notification frequency should prevent alarm fatigue and avoid risk to a child.

18. Technical architecture

Tier	Proposed stack	Responsibilities
Child/parent/admin web	Next.js App Router, React, Tailwind, accessible PWA	Checking/chat interface, parent portal, UCC analytics
Native Android child agent	Kotlin (or supported native module with React Native)	VPN/service status, device-specific permissions, link handling, push alerts
Core API	Node.js/Express or NestJS, JavaScript	Identity, pairing, policies, role-based access, evidence and alerts
Database	PostgreSQL	Accounts, consent, minimal incidents, campaigns, activity state and audit trails
AI service	Python + FastAPI	Text classifiers, risk logic, embeddings, screenshot extraction, evaluation
Vision/LLM provider	Configurable hosted or local vision-capable model	Contextual understanding and child-friendly explanation
Testbed compute	Assigned UCC workstation through SSH	GPU inference/training/evaluation with recorded evidence of actual use
Channel adapters	Telegram, SMS provider, future WhatsApp	Standardise inbound messages into shared engine requests

Logical data flow: Child Android/PWA/Telegram/SMS -> authenticated core API -> AI/rules service -> evidence-based decision -> child explanation; safe alert to guardian if appropriate; minimised event to dashboard. External parties receive nothing until a separate approved referral/export action occurs.

18.1 API families (proposed)

`/auth/*`, `/families/*`, `/children/*`, `/analysis/*`, `/reports/*`, `/safety/support/*`, `/policy/*`, `/devices/*`, `/admin/overview`, `/admin/incidents`, `/admin/campaigns`, `/admin/organisations`, `/admin/exports`, `/integrations/telegram/webhook`, `/integrations/sms/webhook`, `/health`.

18.2 Data entities (minimum)

`User`, `GuardianChildLink`, `ChildProfile`, `ConsentRecord`, `DeviceEnrollment`, `ProtectionPolicy`, `DeviceHeartbeat`, `AnalysisRequest`, `ThreatSignal`, `SanitisedEvidence`, `IncidentReport`, `Campaign`, `ImpersonatedOrganisation`, `ParentAlert`, `ReferralRequest`, `AdminReview`, `ExportAudit`.

No passwords, PINs, raw authentication codes or complete private browsing histories belong in these tables.

19. Recommended datasets, models and evaluation

The official Track 4 guide recommends resources including **PhishTank, URLhaus, OpenPhish, UCI phishing datasets and Hugging Face phishing/spam detection models**; these are examples, not mandatory ingredients. Store their URLs, licence/terms, access restrictions and retrieval instructions in a data registry. Never assume every feed is free for unrestricted redistribution or bulk download.

For local relevance, generate a **small synthetic Ugandan scam test set** in English, Luganda and Runyankole, including fake Mobile Money promotions, school bursaries, OTP theft and impersonated teachers. No real minors' messages or imagery in hackathon fixtures.

Evaluation must separately report **precision, recall, false positives, false negatives**, useful latency measures and observed failure cases. Where feasible, test per-language and distinguish URL-threat detection from content-category filtering. Measure parental-alert delivery separately from model classifications. Avoid invented model metrics.

20. Safeguarding, compliance and trust requirements

The prototype must follow the hackathon's stated safeguarding conditions and applicable Ugandan child and data-protection requirements. This includes privacy by design, purpose limitation, role-specific access, audit logs, restricted evidence access, encryption in transit, secure credential management, a deletion/retention schedule, and documented parental/child notices.

Special treatment of high-risk disclosures: Never make autonomous high-impact decisions based solely on a generative model. Seek expert review of crisis-support language, safe escalation, and limits of confidentiality. External reporting and guardian contact must have a lawful, proportionate safeguarding basis; under-18 status alone does not justify broadcasting confidential conversations.

The user must know whether they are interacting with an AI, whether the mode is a demonstration, what data are collected, when protection is active, and which services are connected.

21. Hackathon MVP (5-6 November 2026)

21.1 P0 - must actually work

Demonstration	Acceptance
Child sends test message/link/screenshot through PWA or Android UI	Request goes to actual API and receives clear status, not decorative response
Genuine AI/ML inference on approved UCC Testbed	Logged test evidence shows real invocation, input, output and workstation used
Risk explanation with threat signals	Correctly labels source/evidence and does not guarantee safety
Guardian-child pairing demo	Valid one-time code establishes a mock or real role link; permissions respected
Parent safety alert	A labelled, testable event arrives in parent view without raw chat content
Admin dashboard	Shows realistic synthetic incidents, sanitized scam messages, impersonated entities and campaigns
Call 116	Opens dialer to 116 on a phone; clearly labelled support resource
Documentation and evaluation	Source, setup, demo video, test results, false positives/ negatives and AI-tool declarations

21.2 P1 - high-value if time remains

- Android default-browser/link-handler proof of concept on a real test phone.
- Android VPN domain-block demonstration using an **owned harmless test domain**, with clear limitations and active status; no guarantee of phone-wide universal coverage.
- Basic English AI companion and parent AI advice interface.
- CSV export of sanitised synthetic intelligence.
- Telegram bot calling the same inference API.

21.3 P2 - roadmap or prototype-only

- Production VPN filtering with reviewed policy lists, DNS-bypass testing and network stability guarantees.
- Device-owner managed app hiding, if feasible under legitimate enrollment.
- Accurate tri-lingual voice agent with photo sharing in live conversation.
- Fully operational SMS/toll-free and WhatsApp channels.
- Automatic referral integration subject to formal partnership and human review.

- Child monitoring and uninstall-state verification under an authorised managed-device framework.

21.4 Suggested live demo

1. A child taps **Is This Safe?** and submits a synthetic fake scholarship message with a test URL.
2. The real AI engine on the Testbed returns technical risk signals and a clear warning.
3. A linked guardian sees a **minimal** safety alert.
4. A second variant of the message appears as a related suspected campaign on the UCC dashboard; display the sanitized scammer message and impersonated institution.
5. A child selects **Something Happened**, receives clear reporting guidance, and can **Call 116**.
6. If Android protection is ready, separately demonstrate one **verified block** versus one permitted test domain.

22. Quality, safety and acceptance tests

Scenario	Expected behavior
Genuine official domain	Does not produce unsupported high-risk assertion
Lookalike/O-versus-zero phishing domain	Warns and explains domain/claimed-brand mismatch
Unknown URL	Returns uncertainty rather than fabricated external verification
Screenshot mixes child and scammer messages	Extracts only relevant evidence; asks for clarification if attribution is unclear
Child selects the wrong mode	Assistant routes to checking or reporting based on described need
Child requests confidential support	Explains privacy and its limits; does not expose full text to routine dashboards
Child indicates imminent danger	Provides supportive urgent guidance and safe human-help route; applies approved escalation
VPN not active	Displays unprotected state, never silently claims all links are blocked
Existing VPN already on	Explains conflict and requests explicit user action
Device check-in missing	States 'not recently seen' rather than 'uninstalled'
Brand impersonated in several reports	Dashboard labels impersonation, not organisational culpability
Export requested	Only authorised users can download a sanitised file with audit trail
Unknown language or unreliable voice transcription	Asks for clarification and offers a safe text fallback

23. Metrics and success criteria

Functional: all P0 screens load, API round-trips work, reports and guardian links survive refresh, and Testbed use is evidenced.
Model: record confusion matrix, precision/recall, false positives/negatives and representative errors; no untested percentages.
Safety: no real identifiable minors' records in demo data; child-confidential information excluded from standard exports.
Operations: clear status for VPN/filtering, AI service outage, last successful check-in and messaging failures.
Usability: child can reach check/report/help in one or two taps; parent sees only actionable, minimal alerts; admin finds scam evidence quickly.

24. Risks, decisions and assumptions

Risk or decision	Product response
Native Android controls larger than hackathon timeline	Build thin Android proof of concept; do not abandon working PWA/dashboard
OS/device privileges are restricted	Separate standard-app capabilities from device-owner scenarios
VPN bypass, encrypted DNS or HTTPS visibility limits	Validate coverage per browser/device; use curated domain filtering and truthful status
Overblocking useful sites	Exceptions, human review and child-help allowlist
AI false positives/negatives	Multiple signals, uncertainty state and measured evaluation
Excessive guardian notification	Minimal alerts, priority thresholds and safeguarding policy
Dangerous confidentiality promises	Explicit policy, trained human escalation, controlled referral
SMS/voice operating costs	Use sandbox/mocks unless funded and approved
Testbed accounts and network availability	Local fallback, SSH/key hygiene, test before demo
Scope expands outside Track 4	Lead with child-focused cybersecurity; frame broader support as extension

25. Immediate build order and team responsibilities

Team A - Frontend/PWA: child check/report/help flows, parent portal, responsive UI, no generic AI-looking design.
Team B - Android: default-link proof of concept and (if possible) one explicit-consent VPN domain-block test; do not claim device owner privileges.
Team C - AI/ML: remote FastAPI on assigned Testbed machine, phishing model + technical URL

features, evaluate false positives/negatives.
Team D - Backend/data: API contracts, role linking, consent, incident events, minimised evidence, dashboard endpoints.
Team E - Intelligence/demo/QA: dashboard campaigns/organisations, sanitised exports, synthetic fixtures, 116 referral, integration testing and three-minute video.

Coordinate through shared API contracts. Run full end-to-end testing before adding new features. Reserve meaningful time before code freeze for the video, documentation, live pitch and contingency.

26. Product vision and final positioning

Chinga's distinctive feature is not that it is 'another AI chatbot'. It is a **layered child-safety ecosystem**:

Prevent suspicious and disallowed links when the platform has a supported control path.
Explain why a message or site may be dangerous.
Support children through private, age-appropriate guidance and human referral.
Involve guardians through proportional, safety-aware permissions and alerts.
Inform authorised institutions about emerging scams through carefully de-identified evidence.

Proposed pitch: '*Chinga helps children check before they click, find help when something goes wrong, and gives families and authorised institutions the tools to respond to emerging online threats - without turning child safety into indiscriminate surveillance.*'

Appendix A. Authoritative technical and service references

- Android `VpnService` and restrictions: https://developer.android.com/reference/android/net/VpnService
- Android VPN service guide: https://developer.android.com/develop/connectivity/vpn
- Android `DevicePolicyManager.setApplicationHidden`: https://developer.android.com/reference/android/app/admin/DevicePolicyManager
- Google Play sensitive-permissions / VPN policy: https://support.google.com/googleplay/android-developer/answer/16558241
- Uganda Ministry of Gender, Labour and Social Development - Sauti 116: https://mglsd.go.ug/uganda-child-helpline-116-2/
- UCC Testbed Hackathon 2026 participant readiness guide and workstation guide (provided by organisers; consult the originals for applicable official requirements).

Implementation note: This PRD is a product proposal, not a claim that Android permissions, referrals, UCC data sharing, APIs, or model evaluation have already been deployed and approved. All prototype datasets involving children must use synthetic or otherwise permitted data.